



ENGINEERING MATHEMATICS-II

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FOURIER TRANSFORMS

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Fourier Sine Transform:



Definition:

The Fourier Sine Transform for the function $f(t)$ is defined by

$$\mathcal{F}_s[f(t)] = F_s(\omega) = \int_0^{\infty} f(t) \sin \omega t dt$$

Definition:

The Inverse Fourier Sine Transform for the function $F_s(\omega)$ is defined by

$$f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega$$

Fourier Sine Transforms:



Problem 1: Obtain Fourier sine Transform of $f(t) = \begin{cases} 4t, & 0 < t < 1 \\ 4-t, & 1 < t < 4 \\ 0, & t > 4 \end{cases}$

Solution:

$$\begin{aligned}\mathcal{F}_s[f(t)] &= \int_0^{\infty} f(t) \sin \omega t dt \\ &= \int_0^1 4t \sin \omega t dt + \int_1^4 (4-t) \sin \omega t dt \\ &= 4 \left[t \frac{(-\cos \omega t)}{\omega} - \frac{(-\sin \omega t)}{\omega^2} \right]_0^1 + \left[(4-t) \frac{(-\cos \omega t)}{\omega} - \frac{(-\sin \omega t)}{\omega^2} (-1) \right]_1^4 \\ &= 4 \left[\frac{(-\cos \omega)}{\omega} + \frac{(\sin \omega)}{\omega^2} \right] + \left[\frac{-(-3\cos \omega)}{\omega} - \frac{(\sin 4\omega - \sin \omega)}{\omega^2} \right] \\ &= \frac{-(\cos \omega)}{\omega} + \frac{(-\sin 4\omega + 5\sin \omega)}{\omega^2}\end{aligned}$$

Fourier Sine Transforms:



Problem 2: Obtain Fourier sine Transform of $f(t) = \frac{e^{-at}}{t}, a > 0$.

Solution:
$$\mathcal{F}_s[f(t)] = \int_0^{\infty} f(t) \sin \omega t dt = \int_0^{\infty} \frac{e^{-at}}{t} \sin \omega t dt = F_s(\omega) \text{ --- (1)}$$

Differentiating wrt ω and applying Leibnitz rule of integration under integral sign, we get,

$$\frac{dF_s(\omega)}{d\omega} = \int_0^{\infty} \frac{e^{-at}}{t} t \cos \omega t dt = \int_0^{\infty} e^{-at} \cos \omega t dt$$

Using the standard result, $\int e^{at} \cos btdt = \frac{e^{at}}{a^2 + b^2} (a \cos bt + b \sin bt)$

$$\frac{dF_s(\omega)}{d\omega} = \left(\frac{e^{-at}}{a^2 + \omega^2} (-a \cos \omega t + \omega \sin \omega t) \right) \Bigg|_0^{\infty}$$

Fourier Sine Transforms:



$$\Rightarrow \frac{dF_s(\omega)}{d\omega} = \left(0 - \frac{1}{a^2 + \omega^2} (-a) \right) = \frac{a}{a^2 + \omega^2}$$

$$\Rightarrow F_s(\omega) = \int \frac{a}{a^2 + \omega^2} d\omega + c = \tan^{-1} \left(\frac{\omega}{a} \right) + c$$

$$\Rightarrow F_s(0) = c$$

By (1), $F_s(0) = 0$

Therefore, $F_s(\omega) = \tan^{-1} \left(\frac{\omega}{a} \right)$

$$\left(F_s(\omega) = \int_0^{\infty} \frac{e^{-at}}{t} \sin \omega t dt - - - (1) \right)$$

Problem 3: Find Inverse Fourier Sine Transform of $\frac{\omega}{1+\omega^2}$

Solution: $f(t) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin \omega t d\omega = \frac{2}{\pi} \int_0^{\infty} \left(\frac{\omega}{1+\omega^2} \right) \sin \omega t d\omega \quad \text{--- (1)}$

$$= \frac{2}{\pi} \int_0^{\infty} \left(\frac{\omega^2}{1+\omega^2} \right) \frac{\sin \omega t}{\omega} d\omega$$

$$= \frac{2}{\pi} \int_0^{\infty} \left(\frac{1+\omega^2-1}{1+\omega^2} \right) \frac{\sin \omega t}{\omega} d\omega$$

$$= \frac{2}{\pi} \left[\int_0^{\infty} \frac{\sin \omega t}{\omega} d\omega - \int_0^{\infty} \left(\frac{1}{1+\omega^2} \right) \frac{\sin \omega t}{\omega} d\omega \right]$$

Fourier Sine Transforms :



$$f(t) = \frac{2}{\pi} \left[\frac{\pi}{2} - \int_0^{\infty} \frac{1}{\omega(1+\omega^2)} \sin \omega t d\omega \right]$$

$$f'(t) = -\frac{2}{\pi} \int_0^{\infty} \frac{\omega}{\omega(1+\omega^2)} \cos \omega t d\omega = -\frac{2}{\pi} \int_0^{\infty} \frac{1}{(1+\omega^2)} \cos \omega t d\omega$$

$$f''(t) = +\frac{2}{\pi} \int_0^{\infty} \frac{\omega}{(1+\omega^2)} \sin \omega t d\omega = \frac{2}{\pi} \frac{\pi}{2} f(t)$$

$$\Rightarrow f''(t) = f(t)$$

$$\Rightarrow (D^2 - 1) f(t) = 0$$

$$\Rightarrow f(t) = C_1 e^t + C_2 e^{-t} \text{ --- (2)}$$

Fourier Sine Transforms :



$$f'(t) = C_1 e^t - C_2 e^{-t} \text{ --- (3)}$$

Put $t=0$ in (2) and (3) we get,

$$f(0) = C_1 + C_2 \text{ and } f'(0) = C_1 - C_2$$

i.e, $C_1 + C_2 = 1 \text{ and } C_1 - C_2 = -1$

On solving , we get, $C_1 = 0 \text{ and } C_2 = 1$

Thus, $f(t) = e^{-t}$



THANK YOU

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